

Product Requirements Document (PRD)

Project 05 — Dual-Core RV32I SoC with Simplified Coherent Memory Subsystem

Document ID: PRD-RISCV-SOC-05
Owner: arafi
Timeline: 5 days
PDK: sky130A (OpenLane)
Bus: AXI4-Lite
FPGA: Arty A7-100T or Zybo Zynq (PL-only)

1. Purpose

This PRD defines **what** the product must do, **who** it is for, and **what conditions** define acceptance. It is deliberately non-implementation — the TRD and Implementation Plan cover *how*.

The product is an educational-grade **dual-core RISC-V SoC** that demonstrates **shared-memory and cache-coherence concepts** through a simplified invalidation-based mechanism. It is not a production chip. It is a proof-of-concept system that a grader can verify in simulation, in physical layout, and on FPGA hardware.

2. Background & Problem Statement

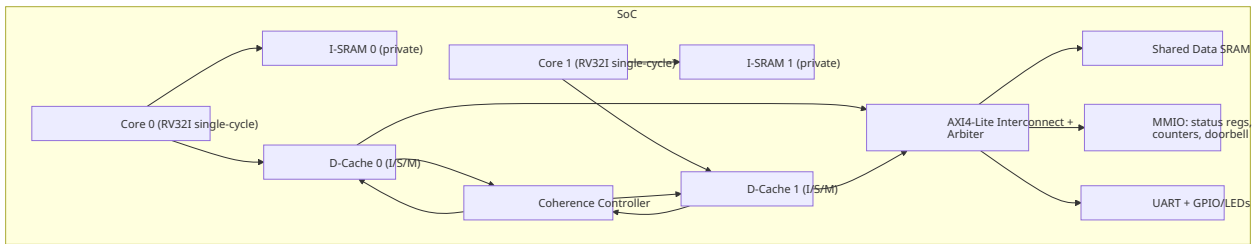
In modern multi-core systems, each processor maintains private cached copies of shared data. When one core modifies shared data, other cores' cached copies become stale. A **coherence protocol** ensures all cores observe a consistent view of memory.

Industry protocols (MESI, MOESI, directory-based snooping) are complex. This product distills the concept to its essence: **two cores, three cache states (I/S/M), and invalidation-on-write** — enough to prove the student understands why coherence is needed and how invalidation makes updated data visible.

3. Target Users & Stakeholders

Stakeholder	Role	What they need from the product
Student (arafi)	Designer & implementer	A working design that passes all verification + physical-design + FPGA gates.
Course evaluator	Grader	Visible proof of concept-demonstration: sim passing, GDS clean, FPGA demo running.
Future reader	Learner	An understandable, documented design showing coherence fundamentals.

4. Product Overview



5. Functional Requirements

FR-1: Dual RV32I Cores

ID	Requirement	Priority
FR-1.1	The SoC shall contain exactly two RV32I base-integer cores.	MUST
FR-1.2	Each core shall be single-cycle (or multicycle — single-cycle preferred).	MUST
FR-1.3	Each core shall execute from a private instruction SRAM.	MUST
FR-1.4	Cores shall be independently resettable to a known PC.	MUST
FR-1.5	Each core shall access shared data through its private data cache, not directly.	MUST

FR-2: Private Data Caches

ID	Requirement	Priority
FR-2.1	Each core shall have a private data cache (direct-mapped, 2–4 lines, 1 word/line).	MUST
FR-2.2	Each cache line shall store tag, data word, and coherence state (I/S/M).	MUST
FR-2.3	Cache shall use write-through policy.	MUST
FR-2.4	On a load miss, the cache shall fetch from shared SRAM and transition I→S.	MUST
FR-2.5	On a store hit/miss, the cache shall write through to shared SRAM and transition →M.	MUST

FR-3: Coherence Subsystem

ID	Requirement	Priority
FR-3.1	The system shall implement a 3-state coherence model (I, S, M).	MUST
FR-3.2	When a core writes a shared line, the coherence controller shall invalidate the other core's cached copy of that line (→I).	MUST
FR-3.3	After invalidation, the invalidated core's next read to that line shall refetch the updated value from shared SRAM.	MUST
FR-3.4	The coherence controller shall maintain per-line state for both cores.	MUST
FR-3.5	The system shall NOT implement full MESI/MOESI or snooping bus.	MUST NOT (by design)

FR-4: Bus & Interconnect

ID	Requirement	Priority
FR-4.1	The system shall use AXI4-Lite for all processor ↔ memory ↔ control communication.	MUST
FR-4.2	An arbiter shall serialize concurrent master requests to the shared SRAM.	MUST
FR-4.3	The arbiter shall use round-robin (or fixed priority) and be deadlock-free.	MUST
FR-4.4	Simultaneous writes to the same address shall produce a deterministic winner.	MUST

FR-5: Memory & Memory Map

ID	Requirement	Priority
FR-5.1	Shared data SRAM shall be a single, unified source of truth.	MUST
FR-5.2	Instruction SRAMs shall be private per core and shall NOT pass through coherence.	MUST
FR-5.3	A memory-mapped region shall expose: coherence status, event counters (invalidations, hits, misses), doorbell, UART regs.	MUST
FR-5.4	Error/unmapped accesses shall return a defined error response (SLVERR/DECERR).	SHOULD

FR-6: Debug & Visibility

ID	Requirement	Priority
FR-6.1	Coherence status and event counters shall be readable via MMIO.	MUST
FR-6.2	UART shall output a human-readable log of coherence events and shared-data values.	MUST
FR-6.3	LEDs shall indicate core activity and coherence events (heartbeat + event pulse).	MUST

6. Non-Functional Requirements

NFR-1: Synthesizability

ID	Requirement	Priority
NFR-1.1	All RTL shall be synthesizable — no initial blocks for logic, no $\$display$ in synthesizable code, no behavioral-only constructs.	MUST
NFR-1.2	No inferred latches. All sequential logic shall use explicit clock + async reset.	MUST
NFR-1.3	SRAM shall be implementable as either OpenRAM macros or register arrays (both must pass synthesis).	MUST

NFR-2: Timing

ID	Requirement	Priority
NFR-2.1	The design shall meet timing at ≥ 20 MHz on sky130A (50 ns period).	MUST
NFR-2.2	On FPGA, the design shall run at ≥ 50 MHz core clock (via MMCM/PLL).	MUST

NFR-3: Verification

ID	Requirement	Priority
NFR-3.1		MUST

ID	Requirement	Priority
	A directed, self-checking SystemVerilog testbench shall cover reset, normal, conflict, and coherence cases.	
NFR-3.2	A UVM environment shall cover concurrent accesses, hit/miss, invalidation, simultaneous writes, reset, and error conditions.	MUST
NFR-3.3	Functional coverage shall track coherence states, events, hit/miss, and error cases.	MUST
NFR-3.4	Code coverage (line/toggle/branch) shall be reported.	MUST

NFR-4: Physical Design

ID	Requirement	Priority
NFR-4.1	OpenLane flow shall run to completion: synthesis → floorplan → power plan → placement → CTS → routing → sign-off.	MUST
NFR-4.2	DRC shall be clean (0 violations).	MUST
NFR-4.3	LVS shall be clean (netlist match).	MUST
NFR-4.4	Reports shall include: area, timing (WNS/TNS setup & hold), power, congestion.	MUST
NFR-4.5	Final layout screenshots shall be captured.	MUST

NFR-5: FPGA Demonstration

ID	Requirement	Priority
NFR-5.1	The design shall synthesize and run on the chosen FPGA board.	MUST
NFR-5.2	The demo shall show: core 0 writes shared data → coherence invalidation → core 1 reads updated value.	MUST
NFR-5.3	UART shall print the before/after values; LEDs shall show event activity.	MUST

7. Scope Boundaries

In Scope

- Two RV32I single-cycle cores.
- Private I-SRAMs, private D-caches (2–4 lines).
- 3-state (I/S/M) invalidation-based coherence.
- AXI4-Lite interconnect + arbiter.
- Shared data SRAM + MMIO debug registers.
- UART + LED status.
- Directed SV testbench.
- UVM verification environment with coverage.
- OpenLane synthesis → GDS on sky130A.
- FPGA demo on Arty A7-100T or Zybo.

Out of Scope (explicitly excluded by the brief)

- Pipelined cores.
- L2 / multi-level cache hierarchies.
- MESI/MOESI or scalable directory/snooping protocols.

- DRAM controllers.
- Interrupts, exceptions, privileged CSRs.
- Zynq PS-side software (if Zybo is used, PL-only).
- Multi-master AXI4 (full AXI4 with bursts).

8. Acceptance Criteria

The project is accepted when **all** of the following are demonstrably true:

#	Criterion	How Verified
AC-1	Both cores execute RV32I programs from private I-SRAM.	Directed TB: core PC advances, correct instruction results.
AC-2	Cache hit/miss behavior is correct.	Directed TB: hit returns cached data, miss fetches from SRAM.
AC-3	Shared-memory reads/writes are correct.	Directed TB: write→read back same value on same core.
AC-4	Coherence invalidation works.	Directed TB: core 0 writes, core 1's line → I, core 1 refetches updated value.
AC-5	Updated data is visible to the other core after invalidation.	Directed TB: core 1 reads new value, not stale.
AC-6	Arbitration handles simultaneous accesses deterministically.	Directed TB: simultaneous writes produce deterministic winner.
AC-7	Reset initializes all state to known values.	Directed TB: post-reset, all caches = I, all regs = 0.
AC-8	Error conditions are handled.	Directed TB: unmapped access returns SLVERR/DECERR.
AC-9	UVM environment passes with functional + code coverage reported.	UVM logs + coverage report.
AC-10	OpenLane flow completes with DRC clean + LVS clean.	OpenLane logs + KLayout LVS report.
AC-11	Area/timing/power/congestion reports are generated.	OpenLane report files.
AC-12	FPGA demo runs and shows coherence via UART + LEDs.	Live board demo or video capture.
AC-13	Brief architecture document is delivered.	Document exists and covers all modules.

9. Deliverables Checklist

- ☐ Synthesizable RTL (all modules)
- ☐ Dual RV32I core integration
- ☐ Instruction/shared SRAM integration
- ☐ Private cache/buffer subsystem
- ☐ Simplified coherence controller
- ☐ AXI4-Lite bus/control interface
- ☐ Coherence status/debug registers
- ☐ UART + GPIO status interface
- ☐ Directed self-checking SV testbench
- ☐ UVM verification environment + coverage reports
- ☐ Timing constraints (SDC)

- ☐ Synthesis report
- ☐ Physical-design reports (area/timing/power/congestion/DRC/LVS)
- ☐ Final layout screenshots
- ☐ FPGA bitstream + demo
- ☐ Brief architecture document

10. Risks & Mitigations

Risk	Impact	Mitigation
RV32I core bugs poison all downstream work.	Critical	Use a known-good single-cycle core; unit-test the core in isolation first.
Coherence "works" but stale register value hides the bug.	High	Design demo SW to re-read after invalidation, not reuse register value.
AXI4-Lite handshake bugs.	High	Write AXI-Lite BFM early; test each channel in isolation.
OpenLane fails on unsynthesizable constructs.	High	Yosys smoke test on day 3; avoid latches, use proper resets.
OpenRAM macro integration eats time.	Medium	Use register arrays for I-SRAM; decide shared SRAM strategy on day 1.
UVM scope creep.	Medium	Directed TB first; UVM layered on top; keep agents minimal.
FPGA timing misses at 50 MHz.	Low	MMCM from board ref clock; retiming/pipelining if needed.

11. Glossary

Term	Meaning
RV32I	RISC-V 32-bit base integer ISA (no compressed, no float, no CSR).
I/S/M	Invalid / Shared / Modified — the 3 cache coherence states.
Write-through	On write, update both cache and shared SRAM simultaneously.
Invalidation	Marking a remote cache line stale (→I) so it must be refetched.
AXI4-Lite	Simplified AXI4: 5 channels, no bursts, single transaction per channel.
OpenLane	Open-source ASIC flow (Yosys + OpenROAD) for sky130.
DRC	Design Rule Check — verifies layout against PDK rules.
LVS	Layout Versus Schematic — verifies layout matches netlist.